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MID-TERMEXAMINATION

(CBCS)

B. Tech (IT/MAE/DMAM), 2nd Semester

(June, 2022)

Subject Code: BEC 110	**Subject: Basic Electrical Engineering**
Time : 1 ½ Hours	**Maximum Marks : 30**

Note: Q. 1 Is compulsory. Attempt any two questions from the rest.

Q1 (2.5*4)

(a) What is the equivalent of 4 capacitors each having value of 9F capacitance connected in series?

(b) Three resistances R, 2R and 3R are connected in delta. Determine the resistances for an equivalent star connection?

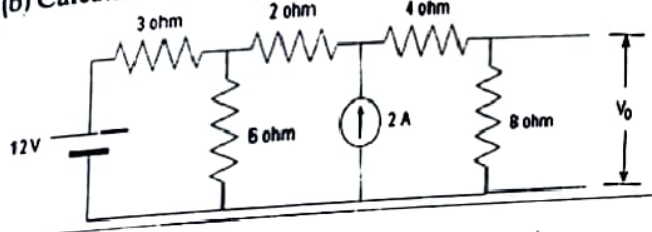
(c) An alternating voltage E=300 sin 314t is applied to a device which offers an ohmic resistance of 30 Ω to the flow of current in one direction, while preventing the flow of current in opposite direction. Calculate Form Factor and Peak Factor?

(d) Explain True Power, Reactive Power and Apparent Power?

Q2 (5,5)

(a) State and Prove Maximum Power Transfer Theorem.

(b) Calculate the voltage V_o by Thevenin's Theorem.



Q3 (5,5)

(a) Explain the Delta to Star conversion and derive the expression of star resistances in terms of Delta resistances.

(b) A series circuit consists of a resistance of 6Ω, an inductance of 500mH and a variable capacitance connected across a 100V, 50Hz supply. Calculate

(i) The capacitance required to produce a series resonance condition

(ii) The voltages generated across both the inductor and the capacitor at the point of resonance.

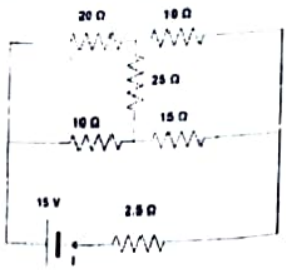
(iii) Bandwidth

(iv) Quality Factor

(v) Power Consumed in the circuit.

Q4 (5,5)

(a) Determine the total current I in the given circuit.



(b) A pure resistance of 50 Ω is connected is in series with a pure capacitor of 100 μF. The series combination is connected across 100 V, 50 Hz supply. Find (a) Impedance (b) Current (c) Power Factor (d) Voltage across R and C.